

Mark Scheme

Q1.

Question	Scheme	Marks	AOs
(a)	(i) Method to find p Eg. Divides $32000 = Ap^4$ by $50000 = Ap^{11}$ $p^7 = \frac{50000}{32000} \Rightarrow p = \sqrt[7]{\frac{50000}{32000}} = \dots$	M1	3.1a
	$p = 1.0658$	A1	1.1b
	(ii) Substitutes their $p = 1.0658$ into either equation and finds A $A = \frac{32000}{1.0658^4} \text{ or } A = \frac{50000}{1.0658^{11}}$	M1	1.1b
	$A = 24795 \rightarrow 24805 \approx 24\,800^*$	A1*	1.1b
		(4)	
(b)	A / (£) 24 800 is the value of the car on 1st January 2001	B1	3.4
	$p / 1.0658$ is the factor by which the value rises each year. Accept that the value rises by 6.6 % a year (ft on their p)	B1	3.4
		(2)	
(c)	Attempts $100000 = 24800 \times 1.0658^t$		
	$1.0658^t = \frac{100000}{24800}$	M1	3.4
	$t = \log_{1.0658} \left(\frac{100000}{24800} \right)$	dM1	1.1b
	$t = 21.8 \text{ or } 21.9$	A1	1.1b
	cso 2022	A1	3.2a
		(4)	
(10 marks)			

(a) (i)

M1: Attempts to use both pieces of information within $V = Ap^t$, eliminates A correctly and solves an equation of the form $p^n = k$ to reach a value for p .

Allow for slips on the 32 000 and 50 000 and the values of t .

A1: $p = \text{awrt } 1.0658$

Both marks can be awarded from incorrect but consistent interpretations of t . Eg.

$$32000 = Ap^5, 50000 = Ap^{12}$$

(a)(ii)

M1: Substitutes their $p = 1.0658$ into either of their equations and finds A

Eg $A = \frac{32000}{1.0658^4}$ or $A = \frac{50000}{1.0658^{11}}$ but you may follow through on incorrect equations from part (i)

A1*: Shows that A is between 24 795 and 24 805 before you see $\approx 24\,800$ or ≈ 24800 . Accept with or without units.

An alternative to (ii) is to start with the given answer.

M1: Attempts $24800 \times 1.0658^t = 32000$ (34)

A1: 24800×1.0658^4 , achieves a value between 31095 and 32005 followed by $\approx 32\,000$ hence A must be $\approx 24\,800$

(b)

B1: States that A is the value of the car on 1st January 2001.

The statement must reference **the car**, its **cost/value**, and **"0" time**

Allow 'it is the initial value of the car' 'it is the cost of the car at $t = 0$ ' 'it is the cars starting value'

B1: States that p is the rate at which the value of the car rises each year.

The statement must reference a **yearly rate** and an **increase in value or multiplier**.

They could reference the 1.0658 Eg "The cars value rises by 6.5 % each year."

Allow " p is the rate the cars value is rising each year" "it is the proportional increase in value of the car each year" "the factor by which the value of the car is rising each year" 'its value appreciates by 6.5% per year' Allow 'the value of the car multiplies by p each year'

Do not allow "by how much the value of the car rises each year" or "it is the rate of inflation"

(c)

M1: Uses the model $100000 = '24800' \times '1.0658'^t$ and proceeds to their $'1.0658'^t = k$

Allow use of any inequality here.

dM1: For the complete method of (i) using the information given with their equation of the model and (ii) translating the situation into a correct method to find ' t '

A1: $(t) = \text{awrt } 21.8 \text{ or } 21.9 \text{ or } \log_{1.0658} \left(\frac{100000}{24800} \right) \text{ oe}$

A1: States in the year 2022. A candidate using a GP formula can be awarded full marks

Allow different methods in part (c).

Eg Via GP a formula

M1: $24800 \times '1.0658'^{n-1} = 100000 \Rightarrow '1.0658'^{n-1} = K$

dM1: Uses a correct method to find n .

A2: 2022

Via (trial and improvement)

M1: Uses the model by substituting integer values of t into their $V = Ap^t$ so that for $t = n, V < 100\,000$ or $t = n+1, V > 100\,000$

(So for the correct A and p this would be scored for $t = 21, V \approx £95\,000$ or $t = 21, V \approx £101\,000$)

dM1: For a complete method showing that this is the least value. So both of the above values

A1: Allow for 22 following correct and accurate results (awrt nearest £1000 is sufficient accuracy)

A1: As before

Q2.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$24 + (6 \times 1.05) + (6 \times 1.05^2)$ minutes	M1	This mark is for a method to find the time taken for the competitor to run 6 km
	$= 96.915$ minutes $= 36$ minutes 55 seconds	A1	This mark is given for finding the total time as required
(b)	For example, 5th km $= 6 \times 1.05^1$ 6th km $= 6 \times 1.05^2$ 7th km $= 6 \times 1.05^3 \dots$ r th km $= 6 \times 1.05^{r-4}$	B1	This mark is given for showing the time taken to run the r th km, as required
(c)	$24 + \sum_{r=5}^{20} 6 \times 1.05^{r-4}$	M1	This mark is given for showing the total time to run the race is the time taken for the first 4 km added to the time taken from 5th to 20th km
	$= 24 + 6.3 \times \frac{(1.05^{16} - 1)}{1.05 - 1}$	M1	This mark is given for using $s = a \left(\frac{1 - r^n}{1 - r} \right)$ where $a = 6 \times 1.05 = 6.3$, $r = 1.05$ and $n = 20 - 4 = 16$
	$= 24 + 149.04$	A1	This mark is given for a correct total time (represented decimally)
	$= 173$ minutes and 3 seconds	A1	This mark is given for finding a correct total time given in minutes in seconds

Q3.

Question	Scheme	Marks	AOs
(a)	Uses $115 = 28 + 5d \Rightarrow d = (17.4)$	M1	3.1b
	Uses $28 + 2 \times "17.4" = \dots$	M1	3.4
	$= 62.8 \text{ (km h}^{-1}\text{)}$	A1	1.1b
		(3)	
(b)	Uses $115 = 28r^5 \Rightarrow r = (1.3265)$	M1	3.1b
	Uses $28 \times "1.3265^4" = \dots$ or $\frac{115}{"1.3265"}$	M1	3.4
	$= 86.7 \text{ (km h}^{-1}\text{)}$	A1	1.1b
		(3)	
(6 marks)			
Notes:			

(a)

M1: Translates the problem into maths using n^{th} term $= a + (n-1)d$ and attempts to find d

Look for either $115 = 28 + 5d \Rightarrow d = \dots$ or an attempt at $\frac{115-28}{5}$ condoning slips

It is implied by use of $d = 17.4$ Note that $115 = 28 + 6d \Rightarrow d = \dots$ is M0

M1: Uses the model to find the fastest speed the car can go in 3rd gear using $28 + 2"d$ or equivalent.
This can be awarded following an incorrect method of finding " d "

A1: 62.8 km/h Lack of units are condoned. Allow exact alternatives such as $\frac{314}{5}$

(b)

M1: Translates the problem into maths using n^{th} term $= ar^{n-1}$ and attempts to find r

It must use the 1st and 6th gear and not the 3rd gear found in part (a)

Look for either $115 = 28r^5 \Rightarrow r = \dots$ o.e. or $\sqrt[5]{\frac{115}{28}}$ condoning slips.

It is implied by stating or using $r = \text{awrt } 1.33$

M1: Uses the model to find the fastest speed the car can go in 5th gear using $28 \times "r^4"$ or $\frac{115}{"r"}$ o.e.

This can be awarded following an incorrect method of finding " r "

A common misread seems to be finding the fastest speed the car can go in 3rd gear as in (a).

Providing it is clear what has been done, e.g. $u_3 = 28 \times "r^2"$ it can be awarded this mark.

A1: awrt 86.7 km/h Lack of units are condoned. Expressions must be evaluated.

Q4.

Question	Scheme	Marks	AOs
(a)	$S_n = a + ar + ar^2 + \dots + ar^{n-1}$	B1	1.2
	$rS_n = ar + ar^2 + ar^3 + \dots + ar^n \Rightarrow S_n - rS_n = \dots$	M1	2.1
	$S_n - rS_n = a - ar^n$	A1	1.1b
	$S_n(1-r) = a(1-r^n) \Rightarrow S_n = \frac{a(1-r^n)}{(1-r)}$	A1*	2.1
		(4)	
(b)	$\frac{a(1-r^{10})}{1-r} = 4 \times \frac{a(1-r^5)}{1-r}$ or $4 \times \frac{a(1-r^{10})}{1-r} = \frac{a(1-r^5)}{1-r}$ Equation in r^{10} and r^5 (and possibly $1-r$)	M1	3.1a
	$1-r^{10} = 4(1-r^5)$	A1	1.1b
	$r^{10} - 4r^5 + 3 = 0 \Rightarrow (r^5-1)(r^5-3) = 0 \Rightarrow r^5 = \dots$ or e.g. $1-r^{10} = 4(1-r^5) \Rightarrow (1-r^5)(1+r^5) = 4(1-r^5) \Rightarrow r^5 = \dots$	dM1	2.1
	$r = \sqrt[5]{3}$ oe only	A1	1.1b
		(4)	

(8 marks)

Notes:

(a)

B1: Writes out the sum or lists terms. Condone the omission of S .

The sum must include the first and last terms and (at least) two other correct terms and no incorrect terms e.g. ar^n

Note that the sum may be seen embedded within their working.

M1: For the key step in attempting to multiply the first series by r and subtracting.

A1: $S_n - rS_n = a - ar^n$ either way around but condone one side being prematurely factorised (but not both)

following correct work but this could follow B0 if insufficient terms were shown.

A1*: Depends on all previous marks. Proceeds to given result showing all steps including seeing both sides unfactorised at some point in their working.

Note: If terms are listed rather than added then allow the first 3 marks if otherwise correct but withhold the final mark.

(b)

M1: For the correct strategy of producing an equation in just r^{10} and r^5 (and possibly $(1-r)$) with the “4” on either side using the result from part (a) and makes progress to at least cancel through by a

Some candidates retain the “ $1-r$ ” and start multiplying out e.g. $(1-r)(1-r^{10})$ and this mark is still available as long as there is progress in cancelling the “ a ”.

A1: Correct equation with the a and the $1-r$ cancelled. Allow any correct equation in just r^5 and r^{10}

dM1: Depends on the first M. Solves as far as $r^5 = \dots$ by solving a 3 term quadratic in r^5 by a valid method – see general guidance or by difference of 2 squares – see above

A1: $r = \sqrt[3]{3}$ or only. The solution $r = 1$ if found must be rejected here.

(b) Note: For candidates who use $S_5 = 4S_{10}$ expect to see:

$$4 \times \frac{a(1-r^{10})}{1-r} = \frac{a(1-r^5)}{1-r} \Rightarrow 4(1-r^{10}) = (1-r^5) \text{ M1A0}$$

$$4r^{10} - r^5 - 3 = 0 \Rightarrow (4r^5 + 3)(r^5 - 1) = 0 \Rightarrow r^5 = \dots \text{ or } 4(1-r^5)(1+r^5) = (1-r^5) \Rightarrow r^5 = \dots \text{ dM1A0}$$

Example for (a)

$$\begin{aligned}
 S_n &= a + ar + ar^2 + ar^3 + \dots + ar^{n-2} + ar^{n-1} \\
 rS_n &= ar + ar^2 + ar^3 + ar^4 + \dots + ar^{n-1} + ar^n \\
 S_n - rS_n &= a(1-r^n) \\
 S_n(1-r) &= a(1-r^n) \\
 S_n &= \frac{a(1-r^n)}{1-r}
 \end{aligned}$$

This scores B1M1A1A0:

B1: Writes down the sum including first and last terms and at least 2 other correct terms and no incorrect terms

M1: Multiplies by r and subtracts

A1: Correct equation (we allow one side to be prematurely factorised)

A0: One side was prematurely factorised